

Agent-Native Research Artifacts (ARA)

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Agent Skills '26 · ACM CAIS Workshop



ORAL

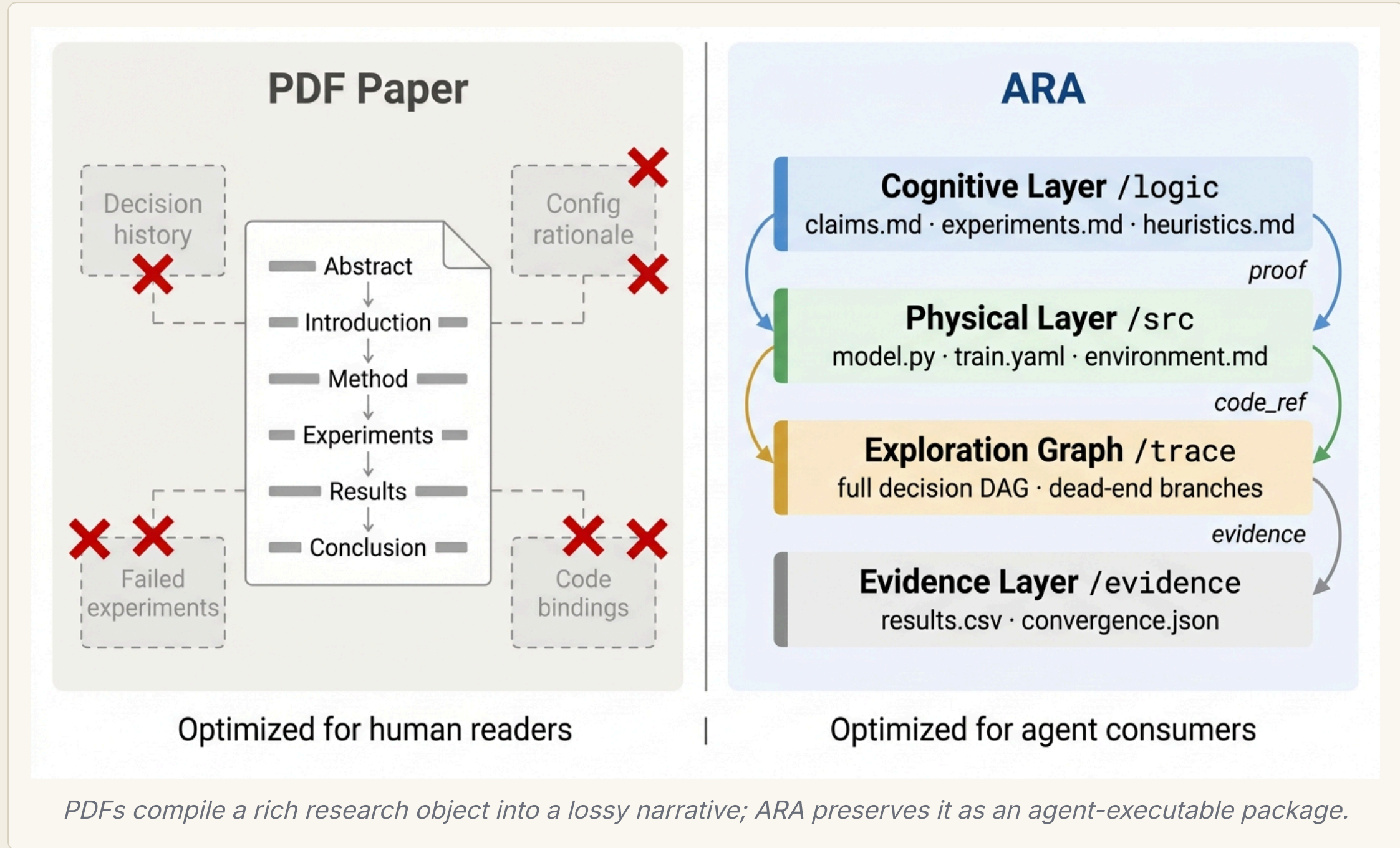


Code & artifacts
github.com/AmberLJC/Agent-Native-Research-Artifact

1. The PDF is Lossy

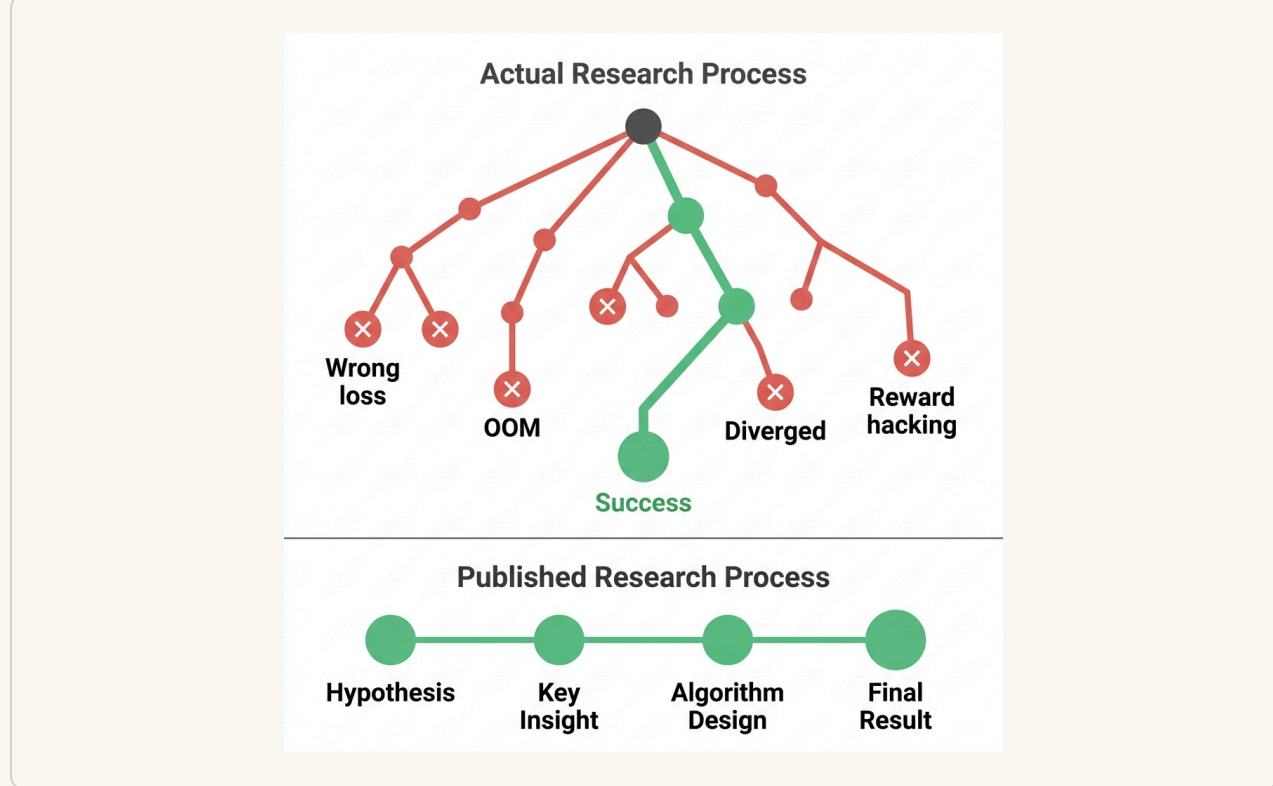
Scientific publication compresses a **branching, iterative research process** into a **linear narrative**, discarding the majority of what was discovered along the way.

Tolerable when every reader was human; **critical** now that AI agents must understand, reproduce, and extend published work.



2. Two Structural Costs of Compilation

Storytelling Tax

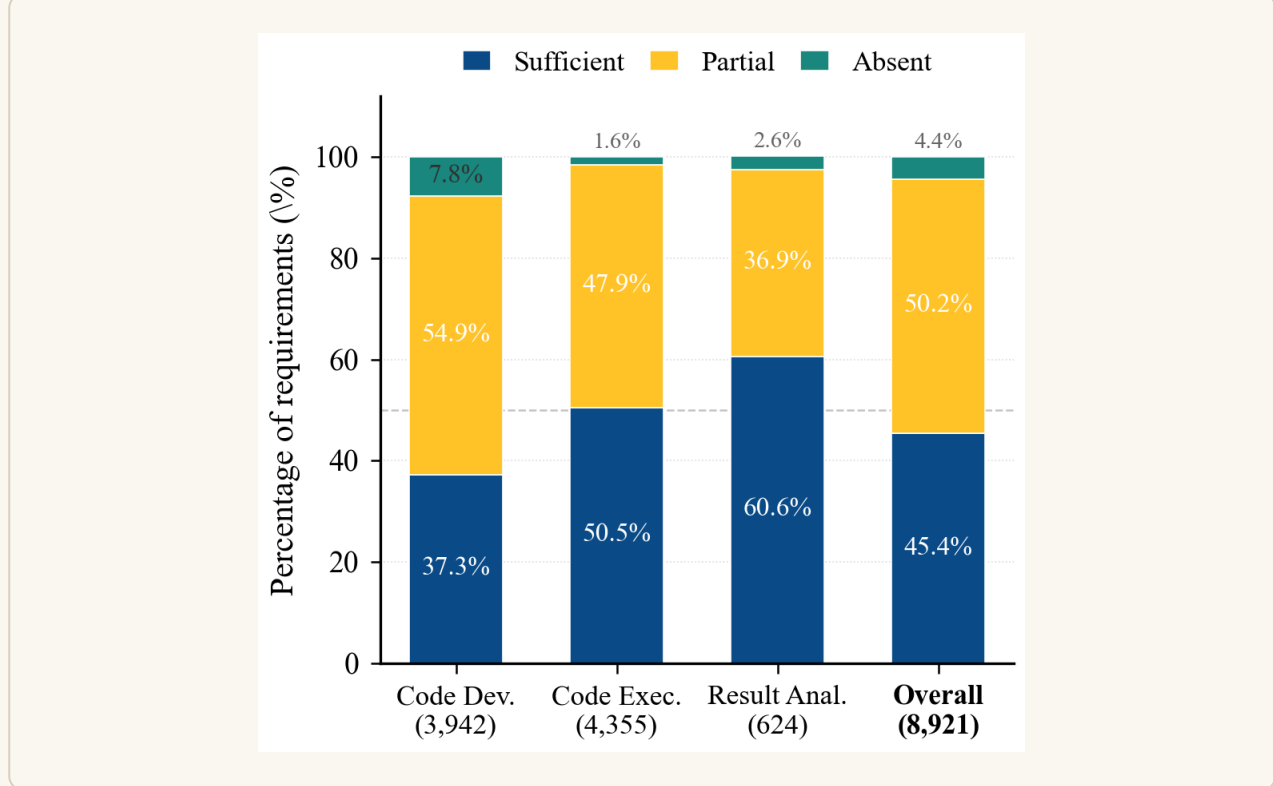


90.2%

\$ ON DEAD ENDS · 24K METR RUNS

Branching exploration collapses into a linear story; agents independently rediscover the same dead ends across groups. Median 113× failed-to-success token ratio.

Engineering Tax



54.6%

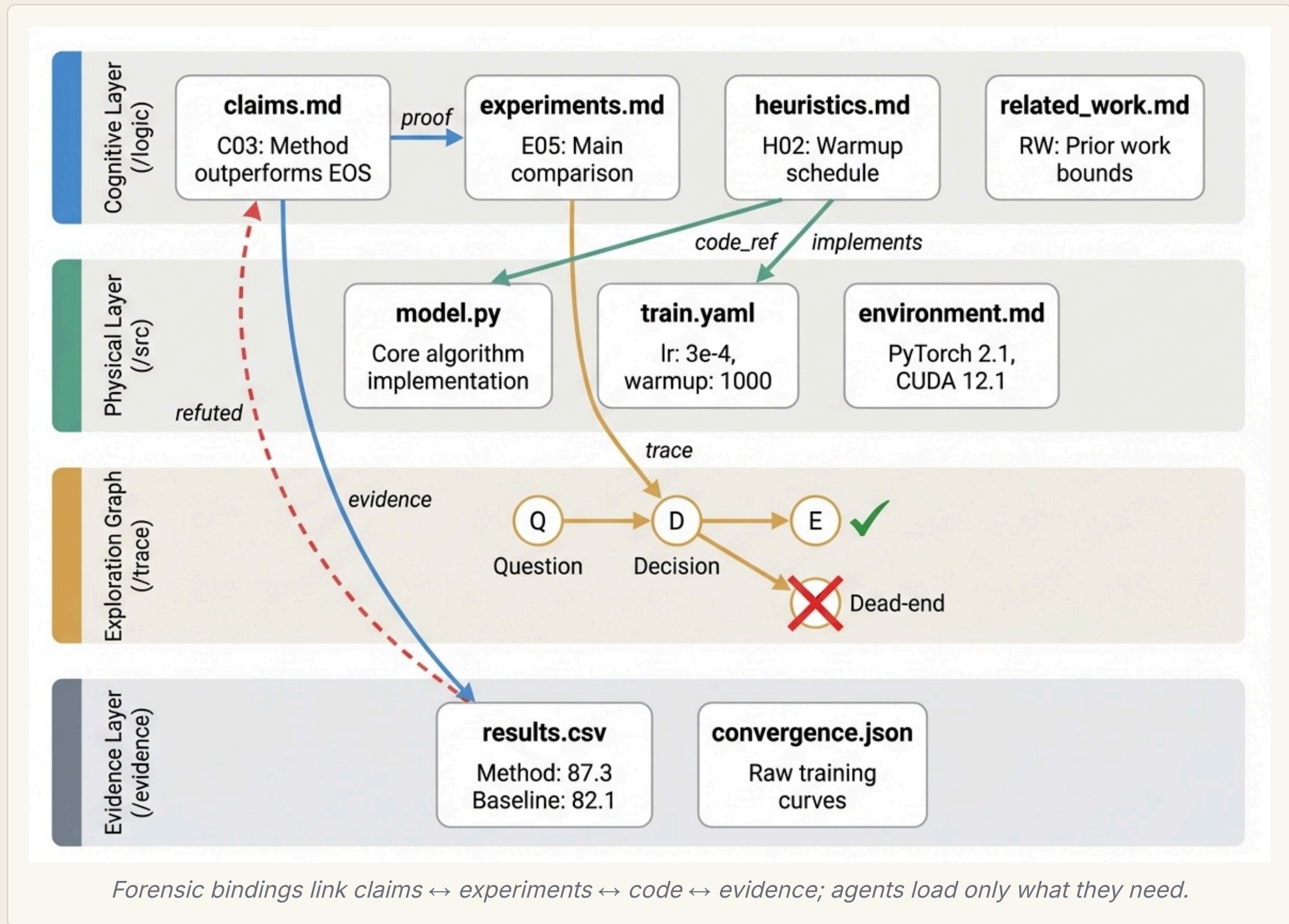
PAPERBENCH REQUIREMENTS MISSING

Between paper and code sits **tacit knowledge** — tricks, configs, decisions written nowhere. Papers convince humans; agents need execution-precision.

3. ARA: A 4-Layer Protocol

A file-system ontology that recasts the research object from *narrative document* to **agent-executable knowledge package**. The paper becomes a compiled view.

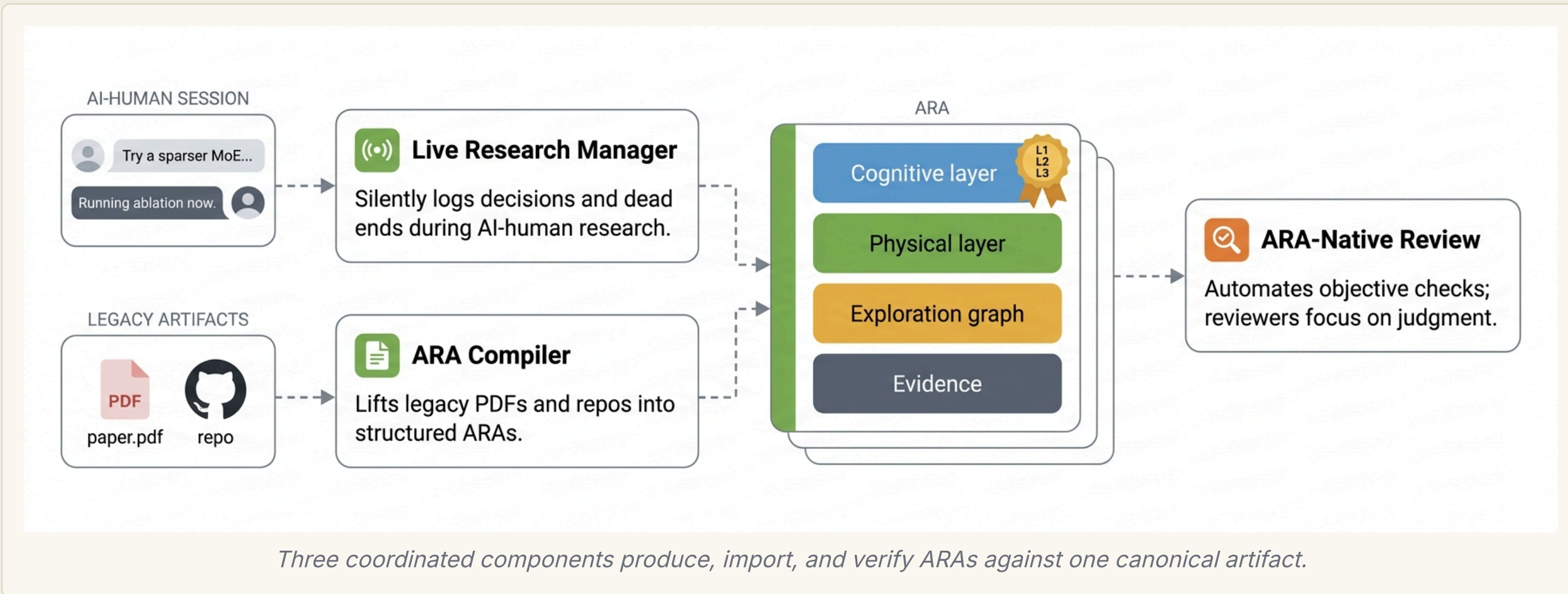
/logic — Cognitive Falsifiable claims, experiments, typed dependency graph. Agent reads first to learn <i>why</i> .	/src — Physical Kernel or repo mode; configs with rationale + bounds; pinned environment. The <i>how</i> .
/trace — Exploration A "git log for research" — every dead end with hypothesis, failure mode, lesson learned.	/evidence — Evidence Raw metrics, logs, curves. Every claim grounded in machine-precise values, never paraphrased.



Principle: Knowledge over Narrative, the organized knowledge is the primary scientific object.

4. The ARA Ecosystem

The protocol's layered structure is too rich to populate by hand. AI agents do the work as a **byproduct of normal research**; the ARA schema is their shared API.



LRM Live Research Manager

Agent skill that **silently crystallizes** ARAs from researcher-agent sessions. Three-stage pipeline at every session boundary: Context Harvester → Event Router → Maturity Tracker.

COMPILER ARA Compiler

Translates legacy PDFs + repos into ARAs via **forensic reconstruction**: Semantic Deconstruction → Cognitive Mapping → Physical Grounding → Exploration Graph Extraction. Converges in ≤3 rounds on all 30 corpus papers.

SEAL ARA Seal — 4 review levels

- L1 Structural**: deterministic, seconds — schema and refs resolve.
- L2 Rigor**: Rigor Auditor scores 6 dimensions (1–5 rubric).
- L3 Reproducibility**: scaled directional checks, evidence-isolated.
- L4 Human Judgment**: the only level not automated — significance, novelty, taste.

5. Evaluation: 3 Research Layers

Does ARA outperform PDF + repo on **understanding, reproduction, and extension** — holding agent, task, and ground truth fixed?

PAPERBENCH · 23 PAPERS	RE-BENCH · 7 TASKS	PROTOCOL
OpenAI's ML-reproduction benchmark: rebuild ICML papers from scratch against 1,743 rubric requirements.	METR's R&D benchmark: build & optimize real ML systems under an 8h cap, scored by a quantitative metric.	Matched ARA vs PDF+repo bundles, identical prompts, paired across all tasks.

① Understanding

450 paired questions

CATEGORY	N	ARA	PDF	Δ
A — Fidelity	300	95.6%	80.8%	+14.8
B — Detail (config)	115	92.6%	67.8%	+24.8
C — Failure knowledge	35	81.4%	15.7%	+65.7
Overall	450	93.7%	72.4%	+21.3

② Reproduction

150 subtasks · 1,743 rubric requirements

64.4% ARA SUCCESS	57.4% PDF + REPO BASELINE	8/5/2 WIN / TIE / LOSS ACROSS 15 PAPERS
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The gap **grows monotonically with difficulty**: easy near-ceiling for both; hard subtasks reveal the configuration gaps PDFs rarely supply. Largest gains on multi-stage training papers (fre, mechanistic-understanding, pinn).

③ Extension

5 RE-Bench tasks · 8h cap · Sonnet 4.6/4.5

Most ambitious claim: **preserving prior failure trajectories** lets the next agent extend the work more effectively. Each ARA is compiled from the official solution + MALT failure traces, filtered so neither bundle ships a worked-out solution.

Early acceleration on every task. ARA reaches a useful first move earlier than the paper baseline on **all 5 tasks**. On `rust_codecontests`, ARA commits to a Rust library at **t = 9 min** after reading heuristic H12; the paper agent finds the same idea only at **t = 395 min**.

ARA wins 3 of 5 under Sonnet 4.6. On the two trailing tasks, the paper agent discovers moves the trace never names — but under **Sonnet 4.5, ARA wins both**. The artifact's value scales with the gap between what the trace documents and what the agent can discover unaided.

ARA replaces narrative compilation with a navigable, executable, verifiable artifact.

+21.3% understanding · +7% reproduction · faster first move on every extension task. The paper becomes a compiled view; the artifact is the primary scientific object.